

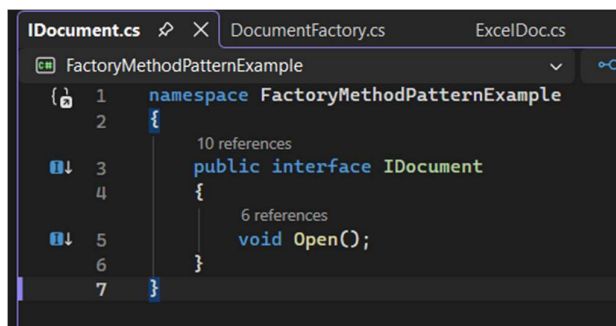
Exercise 2: Implementing the Factory Method Pattern

Scenario:

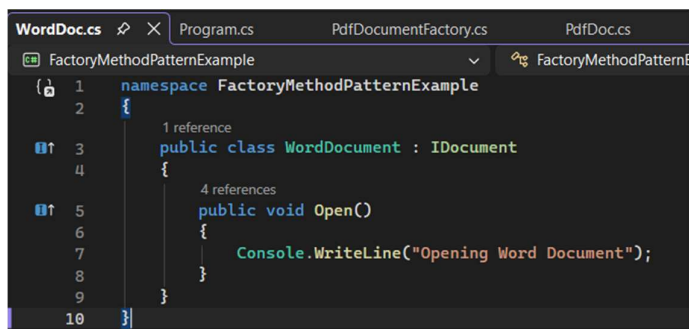
You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

Steps:

1. **Create a New Java Project:**
 - Create a new Java project named **FactoryMethodPatternExample**.
2. **Define Document Classes:**
 - Create interfaces or abstract classes for different document types such as **WordDocument**, **PdfDocument**, and **ExcelDocument**.
3. **Create Concrete Document Classes:**
 - Implement concrete classes for each document type that implements or extends the above interfaces or abstract classes.
4. **Implement the Factory Method:**
 - Create an abstract class **DocumentFactory** with a method **createDocument()**.
 - Create concrete factory classes for each document type that extends **DocumentFactory** and implements the **createDocument()** method.
5. **Test the Factory Method Implementation:**
 - Create a test class to demonstrate the creation of different document types using the factory method.



```
1 namespace FactoryMethodPatternExample
2 {
3     10 references
4     public interface IDocument
5     {
6         6 references
7         void Open();
8     }
9 }
10
```



```
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class WordDocument : IDocument
5     {
6         4 references
7         public void Open()
8         {
9             Console.WriteLine("Opening Word Document");
10        }
11    }
12 }
```

```
PdfDoc.cs  IDocument.cs  DocumentFactory.cs  ExcelDoc.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class PdfDocument : IDocument
5     {
6         4 references
7         public void Open()
8         {
9             Console.WriteLine("Opening PDF Document");
10        }
11    }
```

```
PdfDocumentFactory.cs  WordDocumentFactory.cs  DocumentFactory.cs  ExcelDoc.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class ExcelDocument : IDocument
5     {
6         4 references
7         public void Open()
8         {
9             Console.WriteLine("Opening Excel Document");
10        }
11    }
```

```
PdfDocumentFactory.cs  WordDocumentFactory.cs  DocumentFactory.cs  ExcelDoc.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     6 references
4     public abstract class DocumentFactory
5     {
6         6 references
7         public abstract IDocument CreateDocument();
8     }
```

```
WordDocumentFactory.cs  WordDoc.cs  Program.cs  PdfDocumentFactory.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class WordDocumentFactory : DocumentFactory
5     {
6         4 references
7         public override IDocument CreateDocument()
8         {
9             return new WordDocument();
10        }
11    }
```

```
PdfDocumentFactory.cs PdfDoc.cs IDocument.cs DocumentF
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class PdfDocumentFactory : DocumentFactory
5     {
6         4 references
7         public override IDocument CreateDocument()
8         {
9             return new PdfDocument();
10    }
```

```
WordDocumentFactory.cs DocumentFactory.cs ExcelDoc.cs PdfDoc.cs Program.cs WordDoc.cs ExcelDocumentFactory.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     1 reference
4     public class ExcelDocumentFactory : DocumentFactory
5     {
6         4 references
7         public override IDocument CreateDocument()
8         {
9             return new ExcelDocument();
10    }
```

```
Program.cs PdfDocumentFactory.cs PdfDoc.cs IDocument.cs DocumentFactory.cs
FactoryMethodPatternExample
1 namespace FactoryMethodPatternExample
2 {
3     0 references
4     internal class Program
5     {
6         0 references
7         static void Main(string[] args)
8         {
9             DocumentFactory wordFactory = new WordDocumentFactory();
10            IDocument wordDoc = wordFactory.CreateDocument();
11            wordDoc.Open();
12
13            DocumentFactory pdfFactory = new PdfDocumentFactory();
14            IDocument pdfDoc = pdfFactory.CreateDocument();
15            pdfDoc.Open();
16
17            DocumentFactory excelFactory = new ExcelDocumentFactory();
18            IDocument excelDoc = excelFactory.CreateDocument();
19            excelDoc.Open();
20    }
```

OUTPUT

```
Microsoft Visual Studio Debug
Opening Word Document
Opening PDF Document
Opening Excel Document

D:\DN5-DotNet-FSE-1\Week-1\Design-Patterns-and-Principle
bug\net10.0\FactoryMethodPatternExample.exe (process 919
To automatically close the console when debugging stops,
le when debugging stops.
Press any key to close this window . . .
```